
Modern Application Development 2

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I am Sweta Priya. I am currently pursuing a BS in Data Science from IIT Madras and is at Diploma level. I have a strong interest in Development and Data Science. I am passionate about building scalable and efficient application and explores new technologies to enhance my skill.

Project Description

In my Modern Application Development 2 project, I am focusing on full stack development. I prepared a Trekking Management Application, a role-based access application designed for adventure organizations to manage treks, staff, bookings and trekking statistics, with three roles: Admin, Trek Staff and User (Trekker).

Technologies Used

Backend

- Flask: Used for building the backend of application
- Flask-RESTful style JSON APIs: Used for creating RESTful APIs
- Flask-SQLAlchemy: Used as the ORM
- Flask-JWT-Extended: Used for authentication (JWT based)
- Flask-Caching: Used for caching

Frontend

- Vue.js 3: Used for building the UI
- Jinja2: Used for the backend entry page only
- Custom CSS (design system): Used for styling
- Axios: Used for API communication

Database

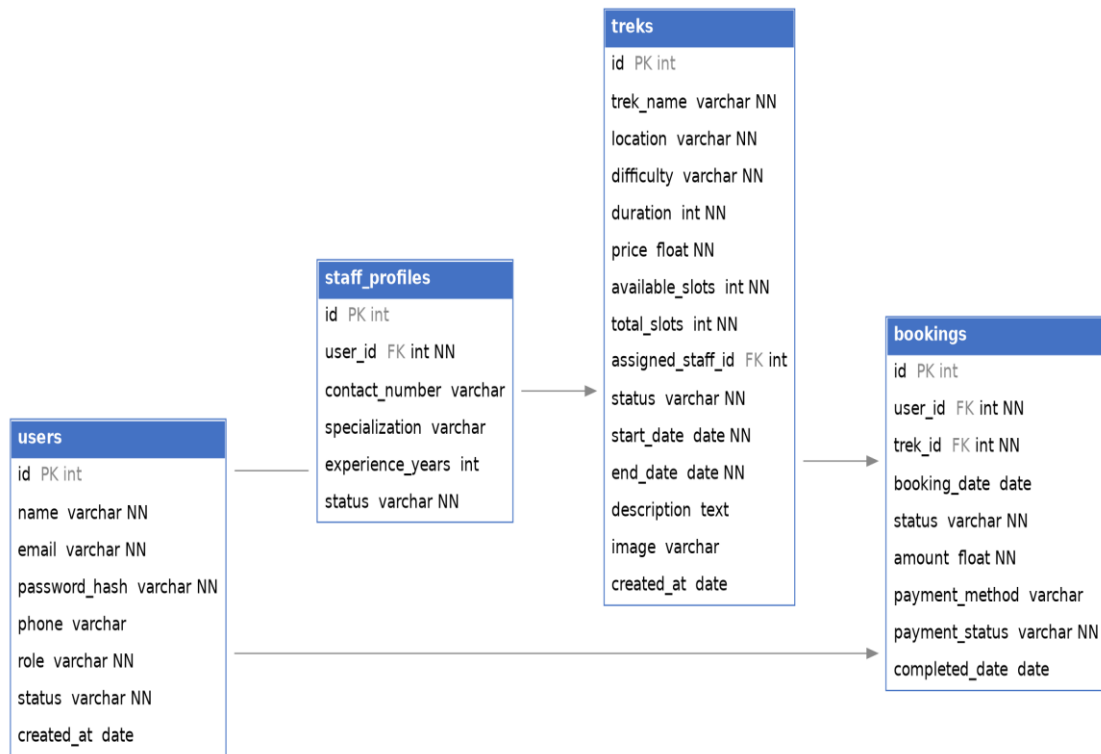
- SQLite: Used as the database

Caching & Background Jobs

- Redis: Used for caching and as the Celery broker/result backend
- Celery (worker + beat): Used for batch jobs and scheduled jobs

- MailHog: Used as a local SMTP catcher for development emails

Database Schema



Routes

- /api/auth/register
- /api/auth/login
- /api/auth/logout
- /api/users
- /api/users/profile
- /api/users/<int:user_id>/status

API

Staff Management

- /api/staff
- /api/staff/<int:staff_id>
- /api/staff/<int:staff_id>/status
- /api/staff/me/dashboard

Trek Management

- /api/treks

- /api/treks/<int:trek_id>
- /api/treks/<int:trek_id>/start
- /api/treks/<int:trek_id>/complete
- /api/treks/<int:trek_id>/participants

Bookings

- /api/bookings
- /api/bookings/<int:booking_id>/cancel
- /api/bookings/export
- /api/bookings/export/<task_id>

Dashboard & Reports

- /api/dashboard/admin
- /api/dashboard/user
- /api/reports

Video Link

Video link: